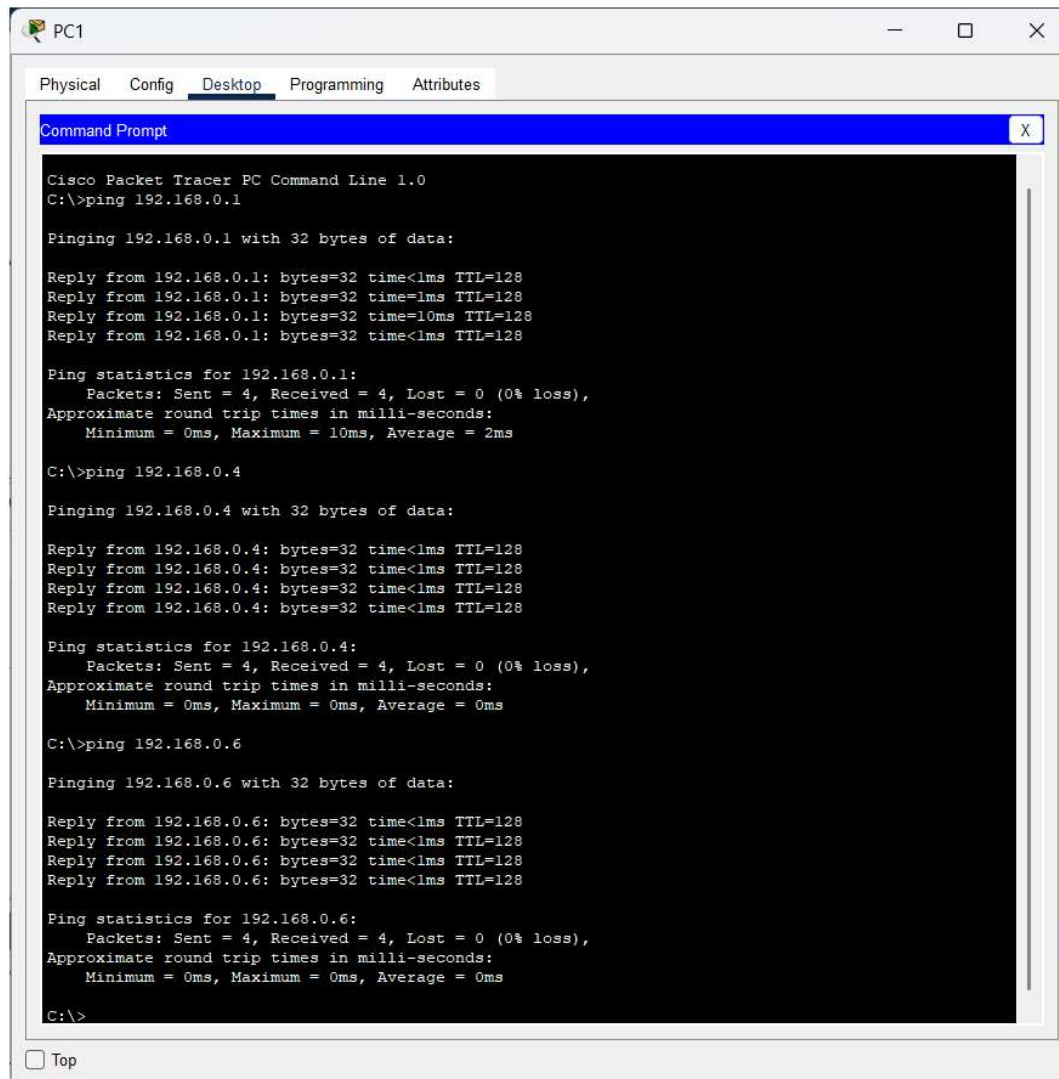


Lab-4

03 February 2026 10:11

1) Construct a tree topology in the Cisco packet tracer, as shown below, and check where the data packets transfer from one system to another.



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time<1ms TTL=128
Reply from 192.168.0.1: bytes=32 time=1ms TTL=128
Reply from 192.168.0.1: bytes=32 time=10ms TTL=128
Reply from 192.168.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:

Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.0.6

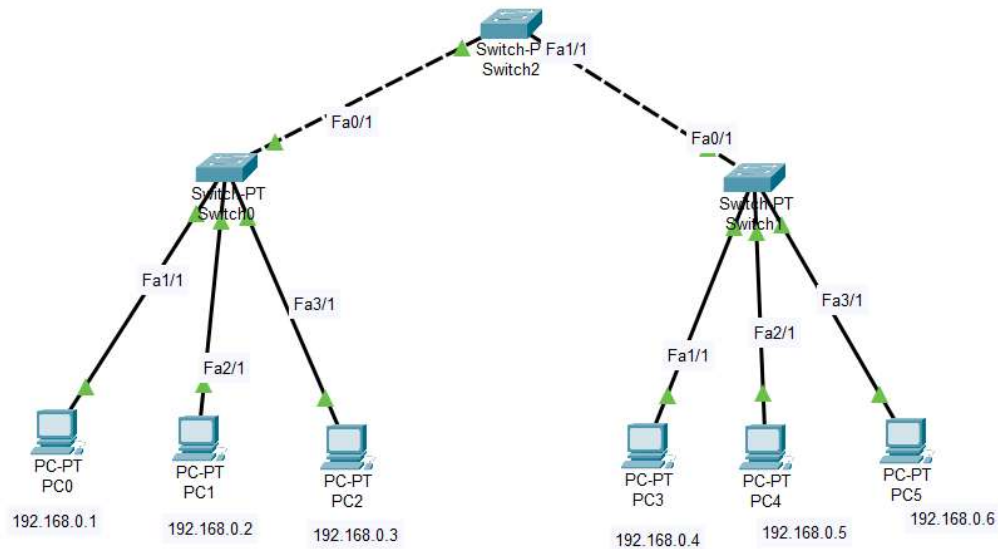
Pinging 192.168.0.6 with 32 bytes of data:

Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128

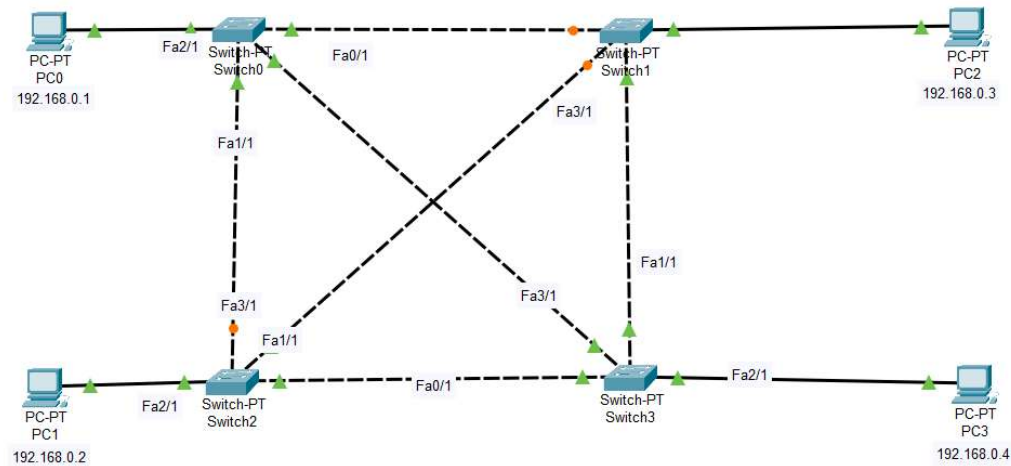
Ping statistics for 192.168.0.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

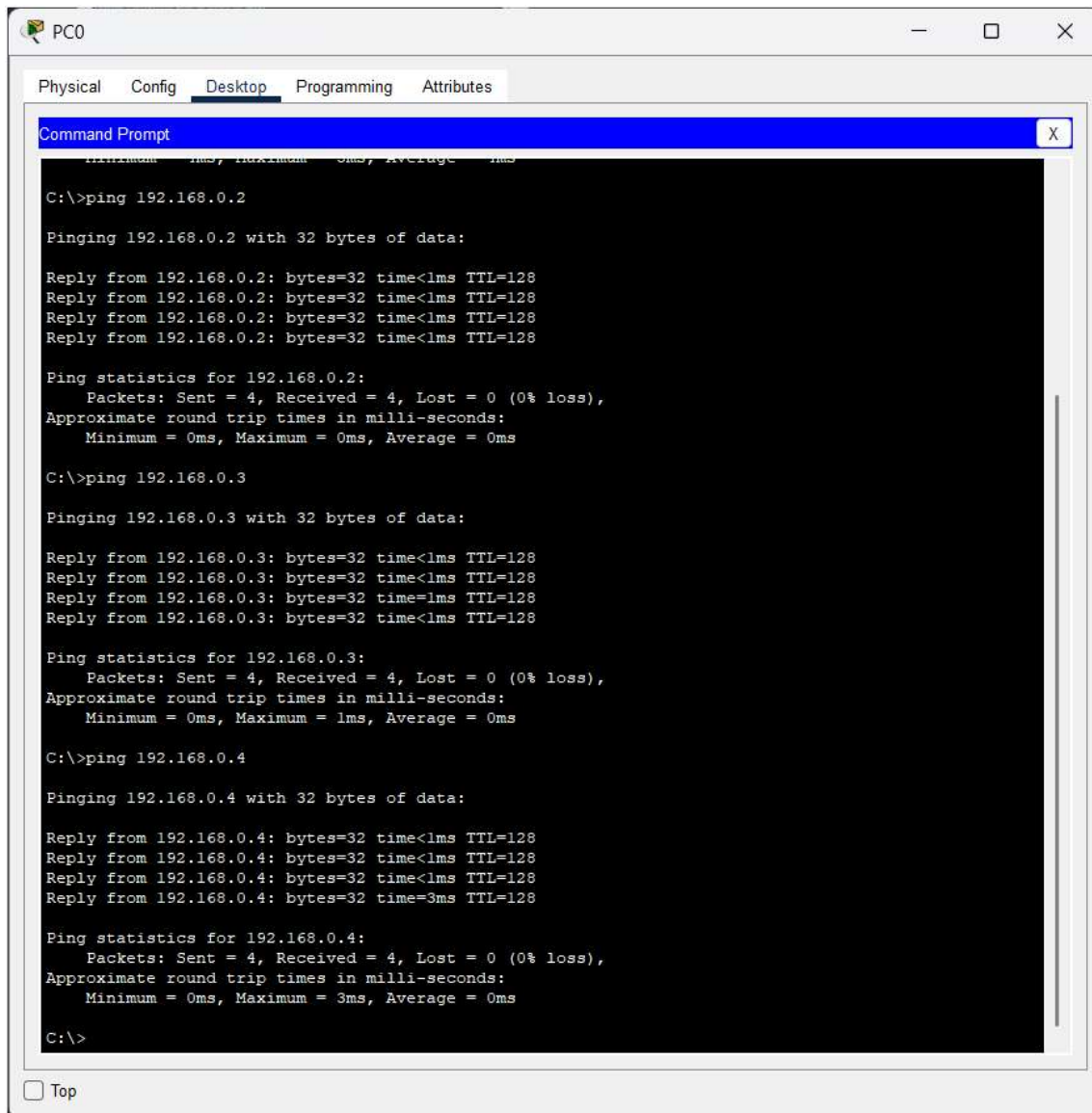
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC5	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC3	ICMP		0.000	N	1	(edit)	(delete)



2) Construct a mesh topology in the Cisco packet tracer as shown below, and check where the data packets are transferring from one system to another.

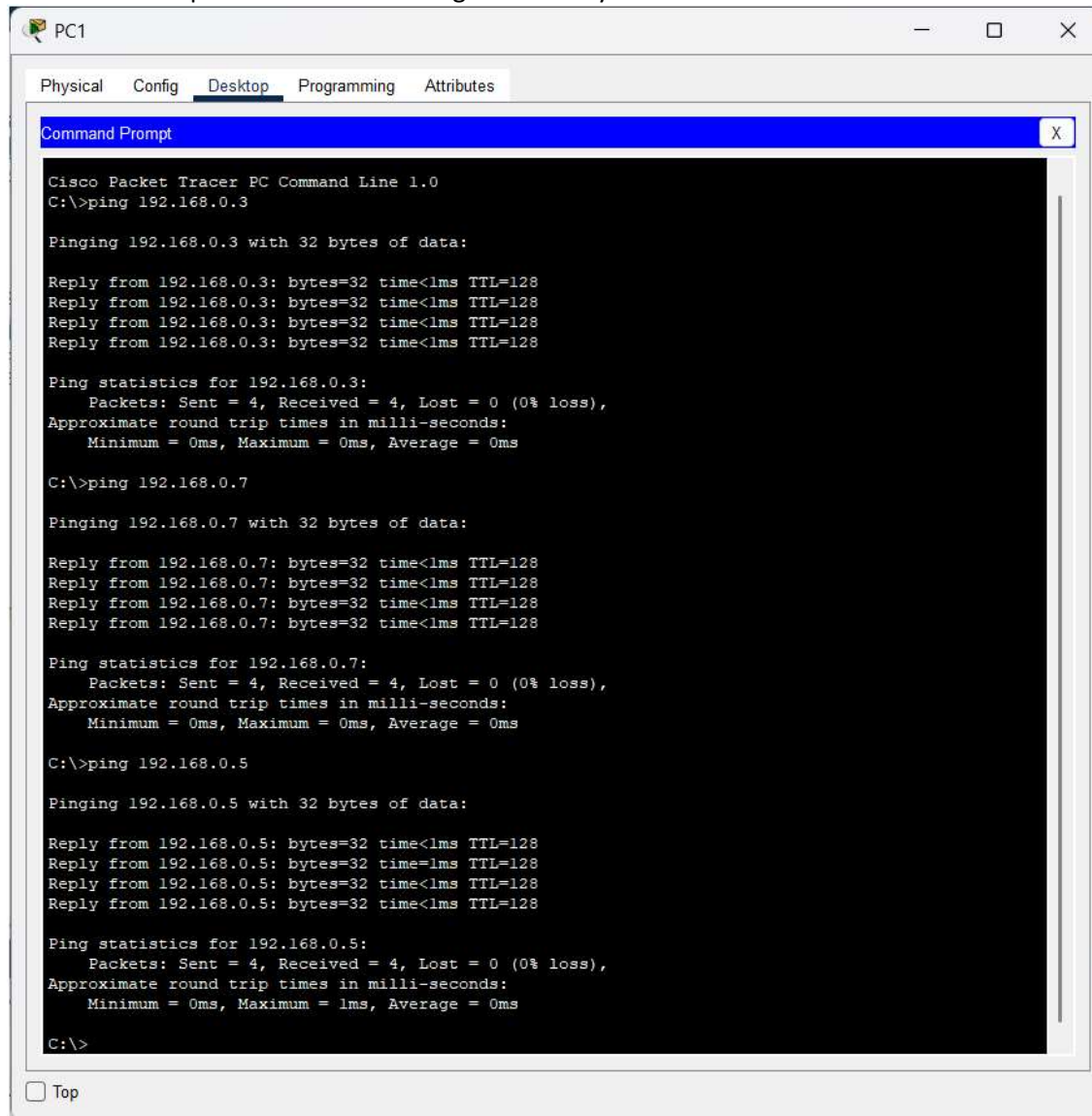


Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC3	PC0	ICMP		0.000	N	2	(edit)	(delete)

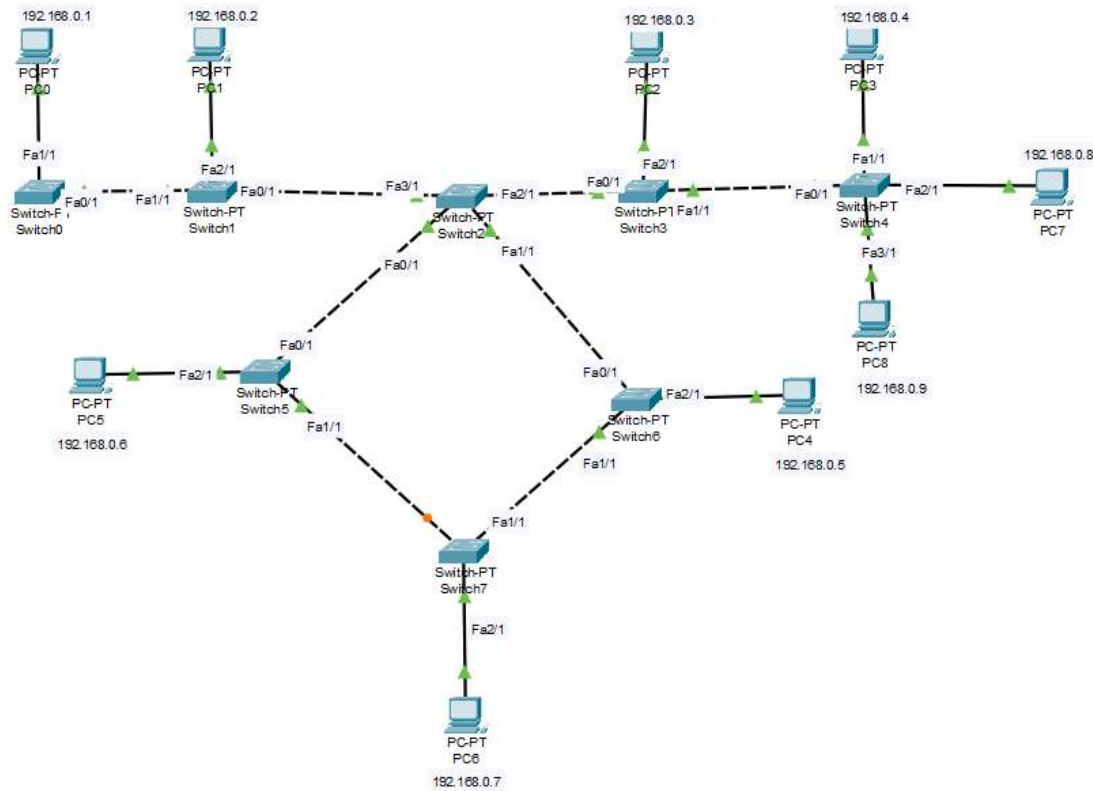


3) Construct a hybrid topology in the Cisco packet tracer as shown below, and check

where the data packets are transferring from one system to another.



	Successful	PC1	PC4	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC5	PC7	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC1	PC6	ICMP		0.000	N	3	(edit)	(delete)



4) In Cisco Packet Tracer, construct a connecting six PCs using two interconnected switches or routers forming two rings. Assign proper IP addresses, ensure full connectivity using the ping command between all PCs, then intentionally disable one link in the ring and observe how data packets are rerouted in the network.

PC7

Physical

Config

Desktop

Programming

Attributes

Command Prompt

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time<lms TTL=128
Reply from 192.168.0.1: bytes=32 time<lms TTL=128
Reply from 192.168.0.1: bytes=32 time<lms TTL=128
Reply from 192.168.0.1: bytes=32 time<lms TTL=128

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:

Reply from 192.168.0.4: bytes=32 time<lms TTL=128
Reply from 192.168.0.4: bytes=32 time<lms TTL=128
Reply from 192.168.0.4: bytes=32 time=17ms TTL=128
Reply from 192.168.0.4: bytes=32 time<lms TTL=128

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms

C:\>ping 192.168.0.5

Pinging 192.168.0.5 with 32 bytes of data:

Reply from 192.168.0.5: bytes=32 time<lms TTL=128
Reply from 192.168.0.5: bytes=32 time<lms TTL=128
Reply from 192.168.0.5: bytes=32 time<lms TTL=128
Reply from 192.168.0.5: bytes=32 time<lms TTL=128

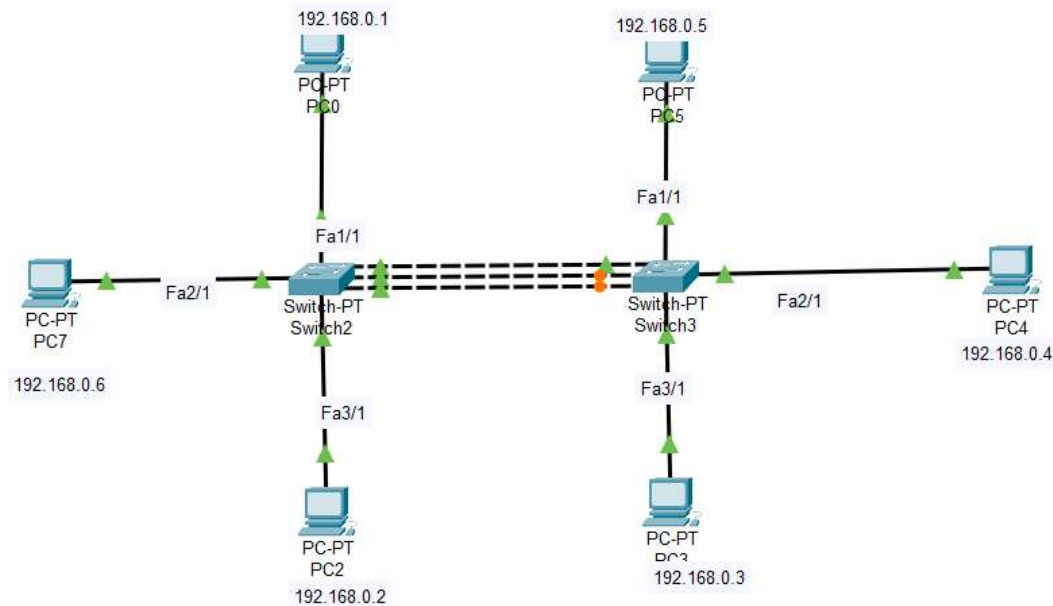
Ping statistics for 192.168.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

☐ Top

	Successful	PC7	PC5	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC2	PC3	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC4	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC7	PC4	ICMP		0.000	N	3	(edit)	(delete)



Step: Intentionally Disable One Link in the Ring

In the given network, the two switches (**Switch2** and **Switch3**) are interconnected using multiple links, forming a **ring topology with redundant paths**. To observe rerouting:

1. One of the **active inter-switch links** between **Switch2** and **Switch3** was intentionally disabled (by unplugging the cable / shutting down the port).
2. Before disabling, data packets were flowing through the **active (green) link**, while the other redundant link was in **blocked state** due to **Spanning Tree Protocol (STP)**.

Observation After Disabling the Link

- As soon as the active link was disabled, the previously **blocked (orange) link automatically became active (green)**.
- The network **did not lose connectivity** between any PCs.
- All PCs were still able to **successfully ping each other**.
- Data packets were **rerouted through the alternate path** available in the ring.

